Listing 10.12 The SsNode::findSplitIndex method

```
function
SsNode::findSplitIndex()
       #1
  coordinateIndex ←
this.directionOfMaxVariance()                                  #2

this.sortEntriesByCoordinate(coordinateIndex)
              #3
  points ← {point[coordinateIndex] for point in
this.getCentroids()} #4
  return minVarianceSplit(points,
coordinateIndex)                    #5
```

#1 Method `findSplitIndex` is defined on `SsNode`. It returns the optimal index for a node split. For a leaf, the index refers to the list of points, while for an internal node it referes to the children's list. Either list will be sorted as a side effect of this method.

#2 Find along which axes the coordinates of the entries' centroids have the highest variance.

#3 We need to sort the node's entries (either points or children) by the chosen coordinate.

#4 Get a list of the centroids of this node's entries: a list of points for a leaf, a list of the children's centroids, in case we are at an internal node. Then, for each centroid, extract only the coordinate given by `coordinateIndex`.

#5 Find and return which index will result in a partitioning with the minimum total variance.